Certified end sets for the exact-rollout gate: an informal proof of `TermSoundR`, and where the straight climb stands

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Abstract

The Lean exact-rollout gate is proved safe against `CitySafety.Safe` (PR #29) under one substantive assumption, `TermSoundR`: an end-of-escape test that passes on a real rollout implies that continuing the escape never crashes. This note reduces `TermSoundR` to *certified end sets*: a dynamics part Dyn (a settled set around the escape law's climb, invariant under the closed loop) and a geometric part Geo (map clearance of a widening funnel). The geometric part, and the reduction, are proved in Lean (PR #37); Sections 2–3 state them in the form proved there.

Section 5 then attacks the dynamics part for the first escape, the *straight climb*, on the real model: `JsbExact`'s equations with fuel frozen, flown by the real escape law, taken over exact reals. Computed with the real plant code: the trim family over 10–1,400 m, a 32-coordinate chart that drops the neutral modes, the reference climb (trim plus its steady lag), the one-frame and one-second linearisations, and the disturbance budget. **The contraction certificate is not obtained.** Three findings explain why, each with numbers: (i) the trim sits exactly on a kink of the frame map (the drag term $0.17\,|\beta|$; the rudder integrator drives β to 0), which a rank-one switched linearisation handles with a large margin; (ii) the per-frame contraction margin is about 10^{-3} (slowest mode 0.137–0.175 s^{-1} at $\Delta t = 1/128$ s), and no per-frame quadratic norm tried contracts on the whole band (best $\lambda = 1.0145$); (iii) the one-second map contracts in one quadratic norm on 600–1,000 m ($\lambda_{\mathrm{lin}} \le 0.981$) but not on 10–450 m (1.26). The linearisation changes because the trim's angle of attack crosses two breakpoints of the c172p's aerodynamic tables as it climbs: the drag table's at $\alpha = 0.0873$ near 375 m and the lift table's at 0.09 near 545 m (Section 5.5; a vertical-wind experiment reproduces either regime at either altitude). A level-off at z_{top} plus a margin would replace everything above z_{top} by one level trim, contracting at 0.93 per second (linear part). So `TermSoundR` for the straight escape remains open; what is proved is the reduction (Theorem 11) to explicit hypotheses S1–S4, with the missing certificate and every other gap listed in Section 7.

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1 Setting

Plant and closed loop. A closed-loop state $x = (\ell, p)$ is the escape law's state ℓ (`CityGate.Esc.Law`) and the aircraft p (`CityGate.Plane`: the `JsbExact` state, the observation after its last frame, and its map anchor). One JSBSim frame ($\Delta t = 1/128$ s) of the escape is the gate's `cityGate.fly`, written F: the law reads the observation and sets the surfaces (`Law.step`), then the plant steps with fuel frozen (`Plane.step`, PR #31).

Exact-real assumption. Throughout, `JsbExact`'s `Float` program is taken to compute its equations exactly over $\mathbb{R}$, with exact maths functions. This gap, and "`JsbExact` = JSBSim", are modelling assumptions; rounding and the C maths library belong to the formal proof (#30). Numerical certificates below are computed in double precision, and each says how its rounding error is controlled.

Map. $\mathcal{S} \subseteq \mathbb{Q}^3$ is the solid set of `CitySafety`: closed building boxes and the region at or below the terrain, with the terrain continued past the map edge. Its highest point is

$$T^\star \;=\; \max(\text{terrain grid}) \;=\; 4633481/5000 \;=\; 926.6962 \text{ m}$$

(exact, from `CityGate/data/terrain.csv`; the highest building top is 458.66 m). The terrain is piecewise linear over the grid and continued by clamping, so its supremum is a grid value.

Crash. For `TermSoundR` the crash shape is the gate's `CrashH h` with $h = 7$: a solid point within 7 m of the reported position in each axis. Via `AirframeNear 7` (a hypothesis of PR #29's theorem) this covers the spec's airframe hull. Write $e = 7$ and $e' = 10$; then $2e^2 = 98 \le 100 = e'^2$, so the horizontal half-diagonal of the e-box is at most e'.

2 End sets and `TermSoundR`

Definition 1 (end set, PR #37 `GateSafety.EndSet`)**.** *A family of end sets indexed by $\theta \in \Theta$ is a pair of predicates* Dyn_θ, Geo_θ *on closed-loop states with (i)* $\mathrm{Dyn}_\theta(x) \Rightarrow \mathrm{Dyn}_\theta(Fx)$*; (ii)* $\mathrm{Dyn}_\theta(x) \wedge \mathrm{Geo}_\theta(x) \Rightarrow \mathrm{Geo}_\theta(Fx)$*; (iii)* $\mathrm{Dyn}_\theta(x) \wedge \mathrm{Geo}_\theta(x) \Rightarrow \neg\mathrm{Crash}(x)$*. Membership is* $\exists\theta.\, \mathrm{Dyn}_\theta(x) \wedge \mathrm{Geo}_\theta(x)$*.*

Theorem 2 (PR #37, `EndSet.termSoundR`)**.** *If every state where the end test passes on a real rollout is a member of some end set, then* ***`TermSoundR`*** *holds.*

Proof. Induction on (i) and (ii) gives membership at every later frame; (iii) gives no crash. □

For the city gate the two families are the helix and the ray funnels below, with Geo independent of the state (so (ii) is trivial). `cityGate_spec_safe_of_tubes` (PR #37) leaves exactly these hypotheses open, per family:

- `hconfR` / `hconfH`: Dyn places the reported position in the funnel's Conf;
- `hinvR` / `hinvH`: Dyn is invariant under F;
- `hterm`: a passing end test implies membership.

Section 4 reduces `hconfR`, `hinvR` and `hterm` for the straight climb to hypotheses S1–S4 on the model; Section 5 reports the attempt to discharge them.

3 The geometric part (proved in Lean, PR #37)

All quantities are rational; distances are compared squared. A reported position is $p = (p_x, p_y, p_z)$ (east, north, up; m).

Definition 3 (ray funnel, `Spec/EndSets.lean`)**.** *A ray funnel R has a start (p_x^0, p_y^0), a nonzero rational direction (a, b) whose length $\|(a,b)\|$ is the unit of along-track and lateral distance, a lateral bound $w \geq 0$ at entry, a widening $s \geq 0$, a climb $g \geq 0$ per unit along-track, an entry altitude z_0 and a release altitude z_{top}. With* $\text{along}(q) = (q_x - p_x^0)a + (q_y - p_y^0)b$, $\text{lat}(q) = (q_y - p_y^0)a - (q_x - p_x^0)b$, $\text{hw}(z) = w + s(z - z_0)$ *and* $\text{reach} = e(|a| + |b|)$*:*

$$\text{Conf}_R(p) :\iff z_{\text{top}} \leq p_z \ \vee\ \big(|\text{lat}(p)| \leq \text{hw}(p_z) \wedge 0 \leq \text{along}(p) \wedge z_0 + g\,\text{along}(p) \leq p_z\big),$$

$$\text{Geo}_R :\iff \forall q \in \mathcal{S}.\ q_z < z_{\text{top}} - e \ \wedge\ \Big(|\text{lat}(q)| \leq \text{hw}(q_z + e) + \text{reach} \wedge \text{along}(q) \geq -\text{reach} \Rightarrow q_z < z_0 + g\max(0, \text{along}(q) - \text{reach}) - e\Big).$$

Definition 4 (helix funnel)**.** *A helix funnel H has centre c, radius r°, radial slack $\delta \geq 0$ at entry, widening s, entry altitude z_0 and release altitude z_{top};* $\text{sl}(z) = \delta + s(z - z_0)$ *and* $d(q) = \|q_{xy} - c\|$*:*

$$\text{Conf}_H(p) :\iff z_{\text{top}} \leq p_z \ \vee\ \big(z_0 \leq p_z \wedge |d(p) - r^\circ| \leq \text{sl}(p_z)\big),$$

$$\text{Geo}_H :\iff \forall q \in \mathcal{S}.\ q_z < z_{\text{top}} - e \ \wedge\ \big(|d(q) - r^\circ| \leq \text{sl}(q_z + e) + e' \Rightarrow q_z < z_0 - e\big).$$

The slack grows with height ($s > 0$) because nothing in the escape laws pulls the aircraft back to a fixed line or circle: the bank and heading holds are open loop in position, so a fixed corridor or annulus is not invariant under persistent model mismatch (rung 1, PR #38, finding 1). Neither Conf asks for a non-negative climb rate; that belongs to Dyn's invariance.

Lemma 5 (clearance, PR #37 `Ray.clear`, `Helix.clear`)**.** *If* $\text{Conf}_R(p)$*,* Geo_R*, $g \geq 0$ and $s \geq 0$, the box $[p \pm e]$ contains no solid point; likewise for the helix with $r^\circ + \delta \geq 0$, $s \geq 0$, $e' \geq 0$, $2e^2 \leq e'^2$. With* ***`AirframeNear 7`*** *the state is not* ***`CrashH 7`*** *(`Ray.not_crashH`, `Helix.not_crashH`).*

Proof sketch. A box point q has $q_z \geq p_z - e$ and moves along-track and laterally by at most reach (or horizontally by at most e'). If $p_z \geq z_{\text{top}}$ then $q_z \geq z_{\text{top}} - e$, which Geo excludes for solid q. Otherwise q is inside the widened corridor at height $q_z + e \geq p_z$, and $q_z \geq p_z - e \geq z_0 + g\max(0, \text{along}(q) - \text{reach}) - e$, which Geo excludes. The Lean proofs are in `Spec/EndSets.lean`. □

Finite check. Geo quantifies over $\mathcal{S}$, but top is piecewise linear on the terrain triangles and there are finitely many boxes, so Geo reduces to finitely many exact rational comparisons: for each terrain cell (the clamped border cells included) and each box footprint, an upper bound on its height and a lower bound on its distance to the corridor or circle. Upper bounds on heights and lower bounds on distances are conservative, hence sound. The release condition $q_z < z_{\text{top}} - e$ for all solid q is $z_{\text{top}} > T^\star + e = 933.6962\,\text{m}$; this note uses $z_{\text{top}} = 934\,\text{m}$. The checker is not written yet (gap G8).

4 The dynamics part: what the model must supply

4.1 Coordinates

Definition 6 (chart)**.** *For a held track ψ_r (clockwise from north), the chart maps a closed-loop state to its altitude h (the reported altitude above sea level) and 32 local coordinates $y \in \mathbb{R}^{32}$:*

1. *attitude: roll ϕ, pitch θ, and heading minus ψ_r (wrapped);*
2. *Earth-relative velocity in track axes (along, right, down);*
3. *body inertial rates (p_i, q_i, r_i) (`pqri`);*
4. *the Adams–Bashforth histories, in track axes: $v_i - v_{i,h0}$, $v_{i,h0} - v_{i,h1}$ (inertial velocity and its two previous values), $\dot{v}_i$ and its previous value $a_{i,h0}$;*
5. *the body angular acceleration `pqridot` and body acceleration `uvwdot` (the latter enters only $\dot{\alpha}$);*
6. *engine rpm and manifold pressure (the lagged `tmap`);*
7. *the law's integrators i_α, i_β, i_ϕ.*

The chart drops the horizontal position and the Earth-rotation angle. Rotations of all inertial vectors about the polar axis are an exact symmetry of the equations (gravity is axisymmetric, the plant's ground is a sphere-relative constant, the atmosphere depends on altitude only), so the east position and the Earth angle are exactly neutral. North position (latitude) and heading are not exact symmetries (Coriolis, J_2); their effect is a disturbance (Certificate 7). The quaternion's normalisation, `JsbExact`'s renormalisation only when $|\,\|q\| - 1| \geq 10^{-10}$, and the stored $\dot{q}$ (a function of q and `pqri`) are covered there too. The remaining state fields are constant in flight with fuel frozen (mass, CG, tanks, engine flags, gear states with no ground contact) or recomputed every frame (the propeller's `act`); the law's clock and held track are constant after the pull-up. The law reads only the observation, and the observation is a function of the state (Certificate 1).

Definition 7 (reference climb, error, settled set)**.** *The reference climb $c(h) \in \mathbb{R}^{32}$ is the chart of the settled straight escape as it passes altitude h (Section 5.2). The error is $\xi(x) = \Sigma^{-1}(y(x) - c(h(x)))$, with Σ the fixed diagonal scaling of `Tools/termsound/trim.py` (`SCALE`). For $Q \succ 0$ and $r > 0$, $D_r = \{x : \xi(x)^\top Q\, \xi(x) \leq r^2,\ h(x) \in [h_{\text{lo}}, h_{\text{hi}}]\}$.*

Centring on the reference climb rather than on the frozen trim $y^\star(h)$ removes the altitude drift of the trim from the disturbance budget to first order: the climb lags the trim by a fixed offset (rung 1, finding 4), and c includes it.

4.2 Hypotheses S1–S4

Fix the straight escape, a held track ψ_r, a ray direction at angle $\psi_r + \theta_{\text{ray}}$, and $N \geq 1$ frames.

Hypothesis 1 (S1, robust N-frame invariance)**.** *$F^N(D_r) \subseteq D_r$ for states whose altitude stays in the band over the N frames. Write $C^* = \bigcup_{k<N} F^k(D_r)$.*

Hypothesis 2 (S2, per-frame bounds on C^*)**.** *There are $\gamma > 0$, $\kappa \geq 0$ and $0 < v_{\min} \leq v_{\max}$ such that for $x \in C^*$ one frame moves the reported position by $\Delta z \geq \gamma \Delta t$, by at most $\kappa \Delta t$ across the ray direction, and by between $v_{\min}\Delta t$ and $v_{\max}\Delta t$ along it.*

Hypothesis 3 (S3, above the release altitude)**.** *From any state of C^* with $z \geq z_{\text{top}}$, the altitude never falls below z_{top} under the escape.*

Hypothesis 4 (S4, airframe)**.** *`AirframeNear 7` on C^* (assumed globally by PR #29's theorem; on C^* the attitude is near level, see Section 5.7).*

Lemma 8 (union of images)**.** *Under S1, C^* is forward invariant.*

Proof. If $x = F^k(y)$ with $y \in D_r$ and $k < N$, then $Fx = F^{k+1}(y)$, which is in C^* when $k+1 < N$, and in $F^N(D_r) \subseteq D_r \subseteq C^*$ when $k+1 = N$. □

Definition 9 (ray dynamics set)**.** *For a ray funnel R with direction (a, b) of length L at angle $\psi_r + \theta_{\text{ray}}$, $w \geq 0$, $s = L\kappa/\gamma$ and $g = \gamma/(L\, v_{\max})$ (both in the funnel's units),*

$$\text{Dyn}_R(x) : \Longleftrightarrow\ x \in C^* \wedge \text{Conf}_R(P(x)),$$

where $P(x)$ is the reported position.

Lemma 10 (`hconfR`, `hinvR`)**.** $\text{Dyn}_R(x) \Rightarrow \text{Conf}_R(P(x))$ *trivially. Under S1–S3,* $\text{Dyn}_R(x) \Rightarrow \text{Dyn}_R(Fx)$.

Proof. $Fx \in C^*$ by Lemma 8. Let $p = P(x)$, $p' = P(Fx)$. *Case $p_z \geq z_{\text{top}}$:* if $p_z \leq h_{\text{hi}}$, S2 gives $p'_z > p_z \geq z_{\text{top}}$; above the band S3 applies. *Case $p_z < z_{\text{top}}$:* by S2 (lengths in the funnel's units are L times metres), $|\text{lat}(p')| \leq |\text{lat}(p)| + L\kappa\Delta t \leq w + s(p_z - z_0) + s\gamma\Delta t \leq w + s(p'_z - z_0)$; $\text{along}(p') \geq \text{along}(p) + Lv_{\min}\Delta t \geq 0$; $p'_z - z_0 \geq p_z - z_0 + \gamma\Delta t \geq g\,\text{along}(p) + gLv_{\max}\Delta t \geq g\,\text{along}(p')$. If instead $p'_z \geq z_{\text{top}}$ the first disjunct holds. □

Theorem 11 (straight climb, conditional)**.** *Let the straight-climb end test be: the escape is past its pull-up, $\xi(x)^\top Q\, \xi(x) \leq r^2$, $h(x) \in [h_{\text{lo}}, h_{\text{hi}}]$, and Geo_R holds for the funnel with start $P_{xy}(x)$, $z_0 = P_z(x)$, $w = 0$, the rational direction nearest to $\psi_r + \theta_{\text{ray}}$ (its angular error added to κ as $v_{\max}$ times the error), and s, g as above. Under S1–S4, the test implies membership of the ray end set, so (with the helix family) `hterm` holds, and `TermSoundR` follows by `cityGate_spec_safe_of_tubes`.*

Proof. At entry lat = along = 0 and $p_z = z_0$, so Conf_R holds, and $x \in D_r \subseteq C^*$. Lemma 10 gives `hconfR` and `hinvR`; Lemma 5 gives the end set's clearance. □

Parametric form, for precomputed street tubes. Nothing above depends on the map except Geo_R: Dyn_R is parametrised by the direction (a, b), the start and z_0, and the constants $r, Q, \gamma, \kappa, v_{\min}, v_{\max}, \theta_{\text{ray}}$ are properties of the law and plant alone. A per-street tube is therefore the pair (Dyn_R with the street's axis as (a, b), an offline exact check of Geo_R for that street's corridor). Two changes make a tube end before the release altitude:

- **Bounded corridor.** Replace Geo_R's quantifier by solids with $\mathrm{along}(q) \le t_\mathrm{h} + \mathrm{reach}$, and Dyn_R by $\mathrm{Dyn}_R \wedge \mathrm{along}(P(x)) < t_\mathrm{h} \vee$ (handover).
- **Handover to a circling end set.** At along-track t_h the law must switch, at a fixed point, to the circling escape whose helix funnel H is pre-certified at that spot. The handover condition: every state of Dyn_R with along $\in [t_\mathrm{h}, t_\mathrm{h} + Lv_{\max}\Delta t)$, flown by the circling law, stays inside a transition corridor that is itself clear (an exact Geo-style check over the transition's reachable positions) and enters H's settled set (Dyn_H, with Conf_H for the helix's entry constants) within a bounded time, with Geo_H checked for that helix. Then $\mathrm{Dyn}_\mathrm{tube} := (\mathrm{Dyn}_R \wedge \mathrm{along} < t_\mathrm{h}) \vee \mathrm{Trans} \vee \mathrm{Dyn}_H$ is invariant. Neither the transition set nor the helix's Dyn_H is certified here (gap G7).

5 The straight climb on the real model

All computations use the real plant and law: `Tools/termsound/TsMap.lean` (the `tsmap` executable) runs `CityGate.Esc.Law.step` and `CityGate.Plane.step` unchanged and only adds the chart, its inverse and a pipe protocol; Python (`Tools/termsound/*.py`, numpy and scipy) does the linear algebra. **Reproduction: `make certify`** prints every number in this section (about 10 minutes on 3 cores).

5.1 The law in its settled regime

After the pull-up ($t \ge 1.5\,\mathrm{s}$ and flight-path angle ≥ 0.03, or $t \ge 6\,\mathrm{s}$) the straight escape holds the track ψ_r it had then: bank target $\phi_T = \mathrm{clamp}(K_\psi\,\mathrm{wrap}(\psi_r - \mathrm{track}), \pm 0.15)$ with $K_\psi = 1$, aileron from $\phi_T - \phi$ with integrator i_ϕ, rudder from $-K_B\beta + K_R(r - g\sin\phi/V) + i_\beta$ with $\dot{i}_\beta = -K_{I,B}\beta$, full throttle, and the cegis elevator law: commanded flight-path gradient $g_c = \mathrm{clamp}(0.15 + 0.02(V - 35), 0.05, 0.35)$, load factor and α commands with $\alpha \le A_\mathrm{max} = 0.24$, and integrator i_α on the α error. Two consequences fix the trim's lateral shape. The integrator i_β forces $\beta = 0$ in any steady state. A steady straight flight needs a small bank $\phi^\star \ne 0$ against the propeller's asymmetry, and $\phi^\star = K_\psi\,\mathrm{wrap}(\psi_r - \mathrm{track})$, so the steady track is offset from ψ_r by $-\phi^\star$: $\theta_\mathrm{ray} = -0.00256\,\mathrm{rad}$ over the whole band, a cross-track drift of $0.088\,\mathrm{m/s}$ relative to ψ_r. Aligning the ray with the steady track (rather than ψ_r) removes this drift from κ.

5.2 Trim family and reference climb

Certificate 1 (observation)**.** *`obsOf` (the observation recomputed from the state, `TsMap.lean`) equals the plant's observation bit for bit on every law-read and position field, over 3,000 frames of a real escape (`tsmap` command `O`; output "obs max bit difference 0").*

Certificate 2 (trim family)**.** *For each h in $\{10, 50, 100, \ldots, 1400\}$ m the frozen trim $y^\star(h)$ (the fixed point of $y \mapsto$ chart after one frame from (h, y), the altitude change dropped) is found by Newton's method with a central-difference Jacobian, continued from a settled climb at 612 m; the residual is below 10^{-8} in scaled units at every grid point (`ref.py`, asserted). Selected values:*

h (m)	V (m/s)	climb (m/s)	γ (rad)	θ (rad)	α (rad)	rpm
10	*34.609*	*4.867*	*0.1411*	*0.2229*	*0.0818*	*2553*
300	*34.558*	*4.824*	*0.1400*	*0.2263*	*0.0862*	*2586*
600	*34.492*	*4.787*	*0.1392*	*0.2297*	*0.0904*	*2621*
900	*34.442*	*4.808*	*0.1401*	*0.2329*	*0.0928*	*2656*
1000	*34.366*	*4.772*	*0.1393*	*0.2332*	*0.0939*	*2662*
1400	*34.009*	*4.592*	*0.1354*	*0.2342*	*0.0988*	*2682*

On the whole band: $\phi^\star = 0.0025$–0.0026, $|\phi_T| = |\phi^\star| \ll 0.15$, $V \geq 34.0 > 30$ *(so the* g_c *clamp, which binds below 30 m/s, does not),* $\alpha \leq 0.099 < A_{\max}$*, and the integrators are inside their clamps (*$i_\alpha = 0.018$, $i_\beta = -0.035$, $i_\phi = 0.114$ *at 612 m). No saturation is active at the trim. The band starts at 10 m because the plant's own ground (its gear) is at sea level and its atmosphere is undefined below it.*

The climb is about 4.8 m/s at a gradient of 0.14 up to 1,400 m. Rung 1 (PR #38, finding 3) reports that the g_c clamp binds above about 2,500 m on this climb: the law then commands a climb it cannot sustain and bleeds speed. This note certifies (when S1 is obtained) only up to $h_{\text{hi}} = 1{,}000$ m, above $z_{\text{top}} = 934$ m, and leaves the regime above to S3 (gap G4).

Reference climb. The reference $c(h)$ is the chart of a real closed-loop climb (smoothed quaternion, see Certificate 7) seeded at 10 m with $y^\star(10)$ plus the lag measured at 612 m, flown for 330 s to 1,405 m, and interpolated linearly in h between frames (37,270 frames). Its distance from the frozen trims is at most 0.16 scaled units (largest in the acceleration histories): the lag. Its residual as a solution, $\|E_h(0)\|$ below, is the altitude-drift disturbance.

5.3 Linearisation and the kink

Let $E_h(\xi)$ be the error after one frame from $y = c(h) + \Sigma\xi$, that is $\Sigma^{-1}(y' - c(h'))$; it includes the dependence of the centre on the new altitude.

Certificate 3 (one-frame Jacobians)**.** *Central differences of* E_h *with steps* 10^{-6} *in scaled units. Error control: steps of* 10^{-4}*,* 10^{-5}*,* 10^{-6}*,* 10^{-7} *give entries agreeing to* 1.5×10^{-5} *(truncation, scaling as the step squared) and* 4.6×10^{-5} *(rounding, scaling as the inverse step) at* 10^{-4} *and* 10^{-6} *against* 10^{-5}*, so the* 10^{-6} *Jacobian is accurate to about* 5×10^{-5} *per entry (`lin.py`). The spectral radius is* 0.99865 *on 10–300 m,* 0.99873 *at 450 m and* 0.99893 *on 600–1,000 m: the slowest mode decays at* 0.173–0.175 s^{-1} *below 450 m and* 0.137–0.138 s^{-1} *above 600 m. At 600 m the slowest rates are* 0.137*,* 0.216*,* 0.244*,* $0.336 \pm 0.794i$*,* 0.456*,* $0.592 \pm 1.905i$ s^{-1}*.*

The trim is on a kink. The drag build-up includes $C_{D\beta} = 0.17\,|\beta|$ (`JsbExact/C172Gen.lean`, `c_CDbeta`), and i_β drives β to exactly 0, so every settled state sits on the switching surface $\beta = 0$ of a piecewise-smooth frame map. One-sided differences disagree there by up to 4.2 scaled units (in the along-track acceleration, from the cross-track velocity column). The map is continuous and piecewise C^1 with two pieces, $\beta > 0$ and $\beta < 0$.

Certificate 4 (the kink is rank one and benign)**.** *At each altitude of the grid* $\{10, 150, 300, 450, 600, 750, 900, 1000\}$ *m the one-sided Jacobians* $A_\pm$ *(at* $\pm 10^{-3}$ *m/s cross-track velocity,* $|\beta| \approx 3 \times 10^{-5}$*, steps* 10^{-6}*) differ by a matrix with singular values* 4.37–4.85 *and then* ≤ 0.022*: a rank-one jump* $uv^\top$*, with* v *essentially* $\nabla\beta$ *and* u *in the along-track acceleration, pitch and vertical acceleration rows. The small-gain value*

$\frac{1}{2}\sup_{|z|=1}|v^\top(zI - A_0/\mu)^{-1}u/\mu|$, *with* $A_0 = (A_+ + A_-)/2$, *is* 0.0146 *at* $\mu = 0.9992$ *and* 0.0135 *at* $\mu = 0.9995$ *(600 m; frequency grid of 4,000 points). Since it is far below 1, the discrete bounded-real lemma gives a common quadratic Lyapunov function for the convex hull* $\{A_0 + t\,uv^\top : |t| \le \frac{1}{2}\}$ *at rate* μ*: the kink alone does not prevent contraction.*

Lemma 12 (mean value for piecewise-smooth maps)**.** *Let* G *be continuous and piecewise* C^1 *on a convex set* K*, with the pieces separated by finitely many smooth hypersurfaces. Then* $\|G(\xi) - G(\zeta)\|_Q \le \sup_{\eta\in K}\|J(\eta)\|_Q\,\|\xi - \zeta\|_Q$*, the supremum over points where* G *is differentiable, for one-sided Jacobians at the surfaces.*

Proof. $t \mapsto G(\zeta + t(\xi - \zeta))$ is Lipschitz on $[0, 1]$, differentiable except where the segment meets a surface (finitely many points, or the segment lies in a surface, in which case a one-sided Jacobian applies), so it is the integral of its derivative. □

So S1 with $N = 1$ follows from $\sup_{D_r}\|J\|_Q \cdot r + \delta \le r$, where δ bounds $\|E_h(0)\|_Q$ plus the disturbances. The supremum must include both pieces at every point.

5.4 Result: no contracting norm found on the whole band

Certificate 5 (per-frame norms)**.** *For* Q *solving* $(A_0/\mu)^\top Q(A_0/\mu) - Q = -W$ *at 600 m, with* $\mu \in \{0.999, 0.9992, 0.9995\}$*,* W *the identity with weight* $\varepsilon \in \{1, 10^{-2}, 10^{-4}, 10^{-6}, 10^{-8}\}$ *on the fast coordinates (Adams–Bashforth histories and accelerations), and* Q *built either at 600 m or summed over the grid, the largest* $\|A_\pm(h)\|_Q$ *over the grid is never below 1. The best,* $\varepsilon = 10^{-4}$*,* $\mu = 0.999$*, built at 600 m (*$\operatorname{cond} Q = 1.1\times10^{10}$*), gives* $\|A_\pm\|_Q = 1.0145$ *(10 m),* 1.0121 *(300 m),* 1.0111 *(450 m),* 0.99908 *(600 m),* 0.99934 *(750 m),* 1.00001 *(900 m),* 1.00068 *(1,000 m) (*`cert.py`*, C2).*

A per-frame certificate has a margin of $1 - \lambda \lesssim 1.1 \times 10^{-3}$ to spend; with $\operatorname{cond} Q \approx 10^{10}$, the Jacobian's variation over the band (and the finite-difference error, about 5×10^{-5} per entry, amplified by up to $\sqrt{\operatorname{cond} Q}$) exceeds it. The two regimes of the slowest mode, below 450 m and above 600 m, are not covered by one norm. Lengthening the step helps:

Certificate 6 (one-second map)**.** *Let* $A_N = A_0(600)^{128}$ *and* Q_N *solve* $(A_N/0.92)^\top Q_N(A_N/0.92) - Q_N = -W$ *(*$\varepsilon = 10^{-4}$*). The frozen one-second products have* $\|A_\pm(h)^{128}\|_{Q_N}$ *of* 0.920/0.924 *(600 m),* 0.931/0.939 *(750),* 0.956/0.962 *(900),* 0.976/0.981 *(1,000), but* 1.201 *(10),* 1.220 *(150),* 1.241 *(300) and* 1.260 *(450 m) (*`nstep.py`*). Sampling the true one-second error map at 600, 900 and 300 m in 80 directions (40 random in the* Q_N *metric and their negatives) gives* $\max\|E_N(\xi) - E_N(0)\|_{Q_N}/\|\xi\|_{Q_N}$ *between* 0.64 *and* 0.80 *for* $\|\xi\|_{Q_N} \le 0.03$ *and about* 2.0 *at* 0.1 *(*`probe.py`*, (a)). At* $\|\xi\|_{Q_N} = 0.03$ *the set allows about* ±2.1 *m/s along-track speed,* ±3.1 *m/s climb rate,* ±0.22 *m/s cross-track speed,* ±0.010 *rad roll and* ±0.009 *rad heading.*

The sampled ratio at 300 m (0.72) is far below the linear bound there (1.24): 80 random directions in 32 dimensions miss the worst direction. This is the reason the document's rules forbid presenting sampled checks as proofs, and it is a warning against the sampled-remainder step proposed for H2: the remainder must be bounded with a covering argument (second-derivative bounds on each piece, or interval enclosures through `JsbExact`'s set-valued `Scalar` interface), not by sampling.

What would close S1. On 600–1,000 m the frozen linear part contracts in Q_N with $\lambda_{\text{lin}} \le 0.981$, leaving a margin of 0.019 per second to cover (a) the switching inside each second (the true one-second Jacobian is a product of mixed pieces; Certificate 4's small gain suggests the effect is small,

but it is not bounded here), (b) the nonlinear remainder over D_r (needs a covering bound), (c) the altitude change over one second (about 4.8 m) and (d) the disturbances below. Below 450 m a second norm is needed, with a handover between the two sets in the overlap, or a norm $Q(h)$ scheduled on altitude with a bound on its change per step. Entries happen low (tens to hundreds of metres), so both regimes are needed.

Certificate 7 (disturbances)**.** *In the per-frame norm of Certificate 5 (`cert.py`, C3):*

- *altitude drift of the reference climb,* $\max_h \|E_h(0)\|_Q$ *over a 1 m grid of 10–1,000 m:* 2.76×10^{-8}*, with neighbouring grid values differing by at most* 3.2×10^{-11} *(a sampled Lipschitz margin in one dimension);*
- *heading and latitude:* $\|E_h(0;\psi_r,N)-E_h(0;0,0)\|_Q \le 3.16\times10^{-3}$ *over headings* $\{45°, 90°, 180°, 270°\}$ *and north offsets* $\{0, \pm30\}$ *km at 10, 500 and 1,000 m. It is dominated by the Coriolis acceleration (*$2\Omega V \approx 5 \times 10^{-3}$ *m/s² vertical when flying east at latitude 0), which enters the acceleration coordinates directly. It is the largest disturbance by five orders of magnitude, and it must be handled by building c and Q per heading, or by moving those coordinates' weight out of Q;*
- *quaternion norm: the plain loop and the loop that rescales the quaternion to unit length before every frame differ by at most* 3.4×10^{-10} *over 60 s. The derivatives above are taken on the rescaled loop, whose frame map has no* 10^{-10} *jumps.*

5.5 What changes between 375 and 550 m: two breakpoints of the α tables

Certificate 8 (the regime change is the trim's angle of attack crossing table breakpoints)**.** *`regime.py` recomputes the frozen trim every 10 m on 300–800 m and records, at each, the per-frame and one-second eigenvalues, the slowest mode's eigenvector, and every discrete quantity one frame reads (`tsmap` command `D`): the segment of each table (lift `CLwbh` and drag `CDwbh` in* α*; the side-force, roll and yaw tables in* α*; propeller thrust and power in the advance ratio J; their Mach corrections in tip Mach; mixture efficiency in fuel/air ratio; ground effect in h/b; the atmosphere's layer), the law's clamps (*g_c*, load factor,* $A_{\max}$*, surface saturations, bank-target clamp, the gain schedule* $\min((50/V)^2, 1)$*) and the stall flag. Over the whole sweep no clamp is active, and only two discrete quantities change:*

- *between 370 and 380 m the trim's* α *crosses* 0.0873 *(*5°*), a row of the drag table `CDwbh`: the slowest rate drops from* 0.1751 *to* $0.1617\,\mathrm{s}^{-1}$*;*
- *between 540 and 550 m it crosses* 0.09*, a row of the lift table `CLwbh`: the slowest rate drops from* 0.1630 *to* $0.1370\,\mathrm{s}^{-1}$ *(one-second factor* $0.850 \to 0.872$*), and the slowest mode changes from one dominated by the vertical acceleration to one dominated by the along-track acceleration (a speed mode).*

Between the breakpoints the rate changes by at most $0.0013\,\mathrm{s}^{-1}$ *per 100 m. J (*0.409–0.419*), tip Mach (*0.77–0.79*) and fuel/air ratio (*0.0825–0.0876*) stay inside one segment of their tables.*

The lift table is piecewise linear: its slope is 5.33 per rad on $[0, 0.09]$ and 10 per rad on $[0.09, 0.10]$, then 4.5 on $[0.10, 0.12]$. The trim's α rises slowly with altitude (thinner air, same weight: 0.0862 at 300 m, 0.0904 at 600 m), so the climb crosses 0.09 near 545 m and the lift curve's slope nearly doubles. That changes how a speed or flight-path error turns into a lift change, hence the speed and flight-path modes. The drag table's breakpoint at 0.0873 does the same to the drag slope near 375 m.

Certificate 9 (controlled experiment: the regime follows α, not altitude)**.** *A constant vertical wind (JSBSim's* `wind` *state,* `tsmap` *command* `W`*) moves the trim's α at a fixed altitude. At 300 m, a 2 ft/s downdraft moves α from* 0.0863 *to* 0.0931*, past both breakpoints, and the slowest rate drops from* 0.1748 *to* $0.1316\,\mathrm{s}^{-1}$*: the high-altitude regime, reproduced at 300 m. At 700 m, a 2 ft/s updraft moves α from* 0.0912 *to* 0.0837*, below both, and the rate rises from* 0.1376 *to* $0.1782\,\mathrm{s}^{-1}$*: the low-altitude regime at 700 m (*`regime.py`*, part 2). The rate is set by which table segments the trim's α is in. (A 6 ft/s downdraft puts the trim at $\alpha \approx 0.135$ and makes it unstable, rate $-0.09\,\mathrm{s}^{-1}$; the certified model has no wind.)*

So the "two regimes" are two more kinks of the same kind as the $|\beta|$ kink, in α instead of β. The trim is not driven onto them, it crosses them as it climbs. Near 375 m and 545 m a settled set D_r contains states on both sides, so the norm must contract both pieces there. Treated like the β kink (`akink.py`, one-sided Jacobians at α 5×10^{-4} either side): the lift breakpoint's jump is dominated by one direction (singular values 437, then 0.60; v along vertical speed, along-track speed and pitch), with small-gain value 0.40 at rate $\rho(A_0) + 3 \times 10^{-4}$. The drag breakpoint's jump has two significant directions (singular values 9.3 and 3.6); its rank-one small-gain value is 0.17, which does not cover the second direction. Each value is below 1, so neither kink alone forbids a common quadratic norm at that altitude. Whether one norm covers all three α segments, the β kink and the whole band together has not been shown (gap G1).

Both breakpoints lie inside the band that must be certified. Entries happen at tens to a few hundred metres, and the clearance funnel is needed up to $z_{\text{top}} = 934$ m. So a climb entering below 375 m crosses both breakpoints before z_{top}, and the band from entry to z_{top} contains all three α segments. The breakpoints are a property of the c172p tables (JSBSim's `c172p.xml`), not of the law. A law with a different climb speed would move them in altitude but not remove them.

5.6 Above z_{top}: a level-off reduces it to one trim

Above z_{top} no clearance is needed; Conf's release disjunct only asks that the altitude stay at or above z_{top} (S3). With "climb forever" that drags in the g_c clamp near 2,500 m and the ceiling near 3,150 m. A level-off removes both. `TsMap.lean`'s `stepHold` is a candidate (not in the gate): the straight escape with the commanded gradient capped, $g_c \leftarrow \min(g_c, \mathrm{clamp}(0.005\,(h_{\text{ref}} - h), \pm 0.35))$, unchanged below $h_{\text{ref}} - 28$ m.

Certificate 10 (level-off trim)**.** *With $h_{\text{ref}} = 1{,}000$ m (z_{top} plus 66 m), the escape flown from the reference climb at 700 m levels off and settles in about 120 s at $h = 1{,}000.20$ m, along-track speed* 70.78 *m/s, $\theta = -0.022$, $\alpha = -0.022$, rpm* 2,909 *(full throttle). Newton's method on the full frame map, with the altitude as a state, finds it as a fixed point (residual $< 5 \times 10^{-9}$). Its slowest per-frame rate is* $0.089\,\mathrm{s}^{-1}$*, so the one-second spectral radius is* 0.9145*. With Q_N from the one-second Lyapunov equation at $\mu = 0.934$, $\|A^{128}\|_{Q_N} = 0.932$ and both one-sided (β) products are at most* 0.934 *(*`leveloff.py`*).*

So a level-off turns the part above z_{top} into one equilibrium, contracting in one norm at 0.93 per second (linear part), with no altitude band and no ceiling. Proving it needs the same covering bound on the remainder as below. It also needs a handover from the climb's settled set to the level trim's, between 972 and 1,000 m, which is above z_{top} and so needs no clearance. Then the band that needs the climbing certificate is entry to 934 m, and both α breakpoints lie inside it.

5.7 Non-vacuity and conservatism

Since r and Q are not certified, the certified test's acceptance cannot be computed yet. As a proxy for the settling cost, `probe.py` (b) flies the real straight escape from 150 m at speeds 25–60 m/s, flight-path angles -0.35 to 0.2 rad and banks 0 to 1.1 rad, and records the first time it is within a box around the reference climb: $|\Delta V| \le 0.5$ m/s, $|\Delta\,\text{climb}| \le 0.5$ m/s, $|\Delta\phi|, |\Delta\psi| \le 0.02$ rad and each integrator within 0.02. Results: from 35 m/s, 4.2–11.6 s; from 25 m/s, 15.1–26.5 s; from 45 m/s, 17.5–19.4 s; from 60 m/s, 25.6–27.5 s. Of the 48 starts, all settle within 60 s and 14 within 16 s; the median is 18.6 s. After entering, every run stayed within about 1.03 times the box for the rest of 60 s. Against the gate's 16 s horizon, starts near the climb speed settle in time and slow or fast starts do not. Rung 1's sets (± 0.11 m/s speed, ± 0.6 m/s climb) are tighter than this box, so these times are lower bounds on the entry time into a rung-1-sized set. On the box the attitude stays within 0.03 rad of level flight at $\theta \approx 0.23$, where the airframe's extent (half-span 5.5 m) is inside the 7 m cube, consistent with S4.

The comparison against today's accepted states (boundary-regimes cases) needs the gate's rollouts with the certified test in place of today's, which needs S1 first (gap G5).

6 Certificates and reproduction

Certificate	Code	Margin or status
1 observation	`TsMap.lean` `0`	exact (0 bits)
2 trims	`ref.py`	residual $< 10^{-8}$; no saturation active
3 Jacobians	`lin.py`, `cert.py` C1	entries $\pm 5 \times 10^{-5}$
4 kink	`cert.py` C1	small gain $0.0146 \ll 1$
5 per-frame norm	`cert.py` C2	**fails**: $\lambda = 1.0145$
6 one-second map	`nstep.py`, `probe.py`	0.981 on 600–1,000 m; **fails** below 450 m
7 disturbances	`cert.py` C3	heading term dominates
8, 9 regimes	`regime.py`, `akink.py`	α breakpoints 0.0873, 0.09
10 level-off	`leveloff.py`	0.932 per second (linear)

Command: `make certify` (builds `tsmap`, creates a venv with numpy and scipy under `.lake/`, runs `ref.py`, `cert.py`, `nstep.py`, `probe.py`, `regime.py`, `akink.py`, `leveloff.py` in `Tools/termsound/`). The document is built with `tectonic docs/termsound-proof.tex`.

7 Gaps

G1 **S1 (contraction) for the straight climb.** Not obtained. Needs one norm that contracts all three α segments of the band (breakpoints at 375 and 545 m, Section 5.5) and both sides of the β kink, or a norm per segment with handovers, a covering bound on the nonlinear remainder on each side of the kink, and the switching inside the N-frame map (Certificates 5, 6).

G2 **Heading disturbance.** 3.2×10^{-3} in the per-frame norm, far above the altitude drift; needs per-heading centres or a norm that does not weight the acceleration coordinates (Certificate 7).

G3 **S2 constants** $\gamma, \kappa, v_{\min}, v_{\max}$ depend on r; without S1 they are not computed. At the trim: climb 4.6–4.9 m/s, along-track speed about 34 m/s, cross-track speed 0 relative to the steady track.

G4 **S3 above the release altitude.** A level-off (Section 5.6) would reduce this to one trim, contracting at 0.93 per second; it is not in the gate's law, and its certificate and the climb-to-level handover are not done. Without it: The climb continues to the g_c-clamp regime (about 2,500 m, rung 1) and a ceiling near 3,150 m (falsifier, #30). Needs either a certified set near the ceiling equilibrium or a terminal level-off in the law; neither is done.

G5 **Conservatism** against today's accepted states needs S1; only the settling proxy (Section 5.7) is reported.

G6 **The 30° and 40° circles** wait for the speed-first law change (#36 step 0); not attempted.

G7 **Tube handover** to a circling end set: condition stated (Section 4), not proved.

G8 **The finite** Geo **checker** on the exported map is not written; $T^\star$ is computed exactly.

G9 **Modelling assumptions**: exact reals for `JsbExact`'s `Float` program, and `JsbExact` = JSBSim.

8 Lessons

1. **Abstract the gate, prove once.** #24 and #25 proved the gatekeeper induction generically; the end-set split keeps that: Theorem 2 and the funnels are generic, the model supplies Dyn.

2. **Exact, conservative runtime tests with a one-time soundness proof** (PR #25's `clearAt`): the funnel check and the $\|\xi\|_Q \le r$ check follow the same pattern.

3. **State assumptions over what the code can actually produce.** Certify-by-construction removes free inputs from the end test.

4. **No special cases at the boundary of the model.** $\mathcal{S}$ is defined everywhere (continued terrain).

5. **Heuristic extrapolation is not a certificate**, and today's helix test fails `TermSoundR` (9 of 2,193 falsifier cases); the circles also settle into a descent at 33–37 m/s (#30 root cause A).

6. **Integral action puts the trim on the kinks it integrates away.** Any $|\cdot|$ in the plant of a quantity a law integrator drives to zero (β here) makes the settled state a switching point. Linearisation-based arguments must be switched from the start.

7. **Per-frame margins are tiny at 128 Hz.** Contraction per frame is $1 - O(\Delta t)$, so the conditioning of Q and the accuracy of derivatives decide feasibility; multi-frame maps restore an $O(1)$ margin at the cost of switching inside the step.

8. **Sampled checks miss worst directions in high dimension** (Certificate 6: 0.72 sampled against 1.24 linear).